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Assignment 01- Project Proposal



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Background

Urban Councils in Sri Lanka provide essential public services to their communities. These responsibilities include handling citizen complaints, issuing permits and licenses, collecting payments for taxes and fines, managing employee salaries, and maintaining council resources and inventory.

Many urban councils, including the Horana Urban Council, still rely on manual, paper-based systems to perform these tasks. Citizens must visit the office in person to submit forms, make payments, or follow up on requests. Council staff keep records in physical files, which is time-consuming and error-prone.

This outdated approach leads to several challenges:

- Slow and inefficient service: Long queues and delays are common.
- No tracking or transparency: Citizens cannot check the status of their requests.
- High risk of errors: Manual data entry can result in incorrect or missing information.
- Limited-service access: Services are available only during working hours.
- Difficult payroll management: Tracking employee leave, overtime, and salary calculations manually are hard.

To overcome these issues, our team is creating a **web-based Urban Council Management System** for the Horana Urban Council. This system will modernize and automate key municipal services using the MERN stack (MongoDB, Express.js, React.js, Node.js), which will improve efficiency and service quality for both citizens and staff.

What is an Urban Council Management System?

An Urban Council Management System is a digital platform that helps local government bodies provide services more efficiently. It connects citizens, officers, and administrators

on one central system, allowing tasks to be completed faster, more accurately, and with greater transparency.

These systems reduce paperwork, manual errors, and allow councils to offer 24/7 access to services such as complaint tracking, online payments, and HR management. As a result, they help build trust between the government and the public.

Our Web Application

Our proposed system, the **Horana Urban Council Smart Management System**, is a complete digital solution with the following main modules:

- Citizen Service Requests: Citizens can submit and track complaints online.
- Permit & License Management: Citizens can apply for approvals digitally; officers can review and approve or reject them.
- Online Payments & Finance Management: Citizens can make secure payments, and financial manager can generate financial reports.
- HR & Payroll System: HR officers can manage salaries, leave, and overtime; employees can view pay slips.
- Inventory Management: Inventory officers can manage assets, update stock, and generate reports.
- User Management: Manage all users efficiently

This platform will ensure ease of use, real-time access, role-based control, and accurate data handling. Although it is designed for the Horana Urban Council, the system can be adapted for use in other local councils across Sri Lanka, supporting the broader goal of digital government transformation.

Problem and Motivation

The Horana Urban Council still relies on paper-based systems for services like handling complaints, issuing permits, collecting payments, and managing employee information. This old method leads to several serious issues.

- Slow and time-consuming services

Citizens must visit the council office in person to submit complaints or make payments. This wastes their time, and sometimes they have to wait in long lines. Council officers also take longer to process requests because everything is done by hand.

- No way to track requests

Once a citizen submits a complaint or application, they cannot check its progress. This causes frustration and confusion, as people do not know when their issues will be resolved.

- Errors and lost information

Paper documents can be lost, damaged, or filed incorrectly. Handwriting mistakes and calculation errors are also common when processes are not handled by a computer system.

- Hard to manage staff information

The HR department struggles to manage employee salaries, leave, overtime, and pay slips. Salary calculations are completed manually, which can lead to delays, disputes, and mistakes.

- Lack of transparency

Since everything is done manually, it is hard to keep clear records and provide proof of work completed. This reduces trust between citizens and the council.

- Limited access

Services are only available during working hours and only at the physical office. People with busy schedules or those who live far away find it difficult to access services.

These issues indicate that the current system is outdated, inefficient, and not user-friendly for citizens or employees.

Motivation

To address these issues, the Smart Horana Urban Council Management System aims to offer a centralized, user-friendly web application that modernizes key municipal operations. By introducing an integrated platform that connects citizens, officers, administrators, and employees, the system will enable:

- Faster and more transparent service delivery
- Secure online payments and digital receipt generation
- Real-time tracking of requests and applications
- Automated salary and HR management
- Efficient processing of permits and licenses
- Easy access to reports and dashboards for decision-makers.

This initiative is not just about automation; it is about empowering local governance to better serve the community. It will reduce paperwork, minimize errors, and give citizens a smoother and more reliable experience when dealing with their local government. By embracing digital transformation, the Horana Urban Council will take an important step toward e-governance, leading the way for smarter cities and stronger community trust.

Aim

This project will design and implement a Smart Urban Council Management System for the Horana Urban Council using modern web technologies. The system will digitize and improve essential municipal operations, such as citizen complaints, license processing, online payments, HR and payroll management, and inventory tracking. By introducing automation and centralized management, the platform will improve service efficiency, transparency, and accessibility for both citizens and council staff.

Objectives

To achieve this goal, the project will focus on the following key objectives:

- **Develop a Centralized Web Platform**

Build a secure, scalable system using the MERN stack (MongoDB, Express.js, React, Node.js) to simplify urban council operations.

- **Citizen Complaint Management**

Implement an online portal for citizens to submit, track, and receive updates on service requests and complaints.

- **Digital Permit & License Processing**

Enable citizens to apply for permits and licenses online while allowing officers to review, approve, and manage applications easily.

- **Online Payment Integration**

Provide a secure payment gateway for taxes, fines, and fees, with automated receipt generation for users.

- **HR & Payroll Automation**

Develop a module to manage employee salaries, leave requests, and overtime calculations with set business rules.

- **Inventory & Asset Tracking**

Keep real-time records of municipal assets, monitor stock levels, and generate inventory reports.

➤ **Role-Based Access Control**

Ensure system security by restricting access based on user roles, such as admin, staff, and citizens.

➤ **Multilingual & Mobile-Friendly Access**

Support multiple languages and make sure the system works well on all devices for 24/7 accessibility.

➤ **Data-Driven Reporting**

Generate financial, HR, and operational reports to help the council make informed decisions.

➤ **Testing & Validation**

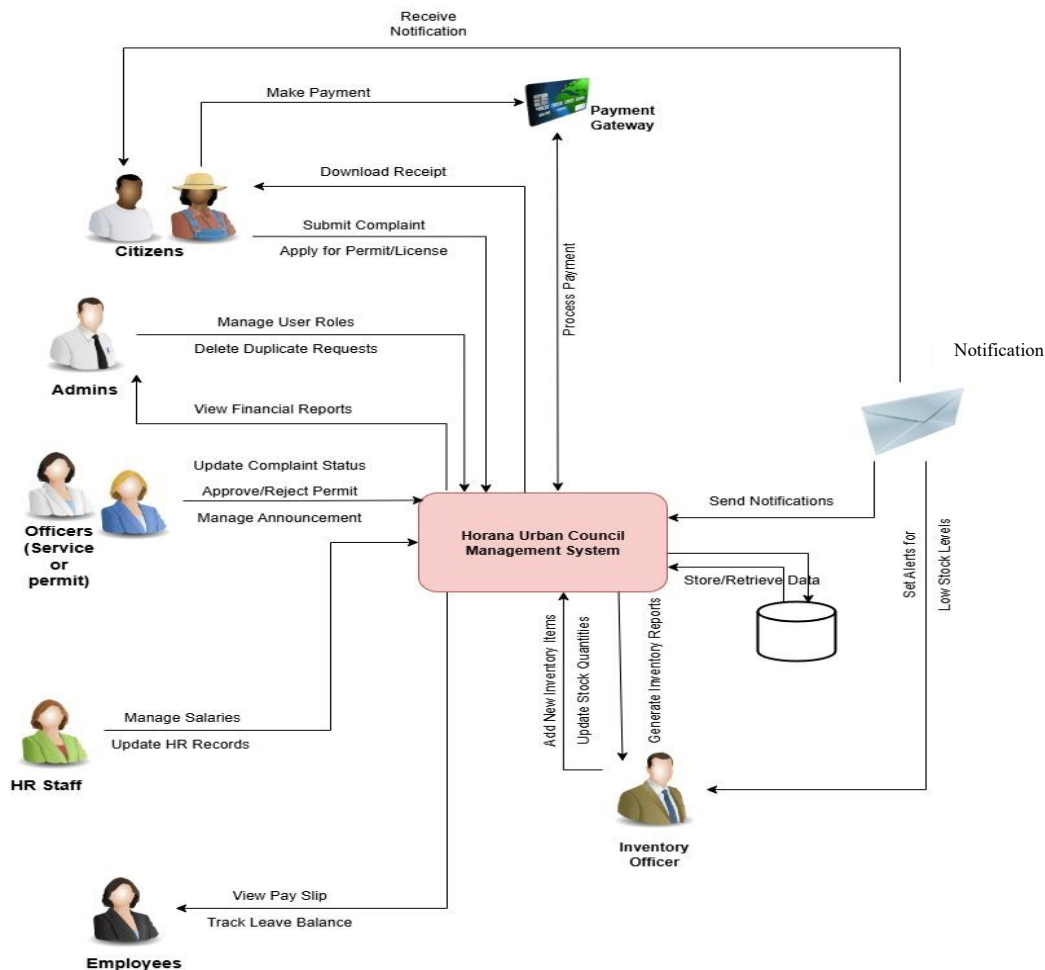
Conduct thorough testing to ensure reliability, security, and user satisfaction before launching.

System Overview

The Horana Urban Council Smart Management System is a complete, web-based platform developed to improve the efficiency, transparency, and accessibility of municipal operations. It is built using the MERN stack, MongoDB, Express.js, React.js, and Node.js, which offers a powerful combination of tools for building scalable and high-performance web applications.

This system is designed to replace the traditional paper-based methods currently used by the Horana Urban Council with a modern digital solution. By automating core processes such as complaint handling, permit and license applications, online payments, HR/payroll management, and inventory tracking, the platform will reduce manual effort, minimize errors, and increase service delivery.

System Diagram



Functional Requirements

1. Citizen Service Request Management

The aim of the Citizen Service Request Management module is to facilitate communication between the Urban Council and the citizens of the Horana region concerning issues. via this system, citizens may submit complaints regarding municipal services like water supply, street lighting, road maintenance, and garbage collection. The simple interface ensures that complaints are properly recorded by supporting them in selecting the issue category and providing the required information. Once a complaint has been filed, citizens can follow its progress in real time, from filing to resolution. This improves handle transparency and trust.

Service officers have access to all complaints that have been filed on their dashboard from an administrative perspective. They may organize complaints according to their urgency, update statuses like "Pending," "In Progress," or "Resolved," and delegate responsibility to the suitable departments. To maintain accurate and clean data, officers can also eliminate complaints that are duplicated or invalid. This module guarantees prompt and thoroughly documented responses while significantly reducing the manual labor typically needed to handle public issues. Additionally, it assists the council in monitoring the frequency and nature of complaints, which can guide future resource allocation and planning. The Urban Council increases operational effectiveness and citizen satisfaction by digitizing this process, which also makes municipal services more responsive and accountable.

2. Permit and License Management

Applying for official documents that include construction permits, business licenses, or event approvals is made easier for citizens by the Permit and License Management module. This procedure used to involve numerous trips to the council office, the submission of paper forms, and lengthy approval periods. By enabling users to apply for

permits using online forms that automatically adapt based on the selected category, this digital module eliminates those issues. To ensure that the correct information is gathered without confusion, a construction permit form, for instance, might request different documents than a business license.

Authorized officers can use the system to examine, approve, reject, or request additional information after an application has been submitted. If required, each officer can submit the application to another department or include notes inside. Any time the status changes whether it's approval, rejection, or a request for more information citizens are automatically notified via email. This retains the applicant updated at every stage and reduces the need for follow-up phone calls or visits.

Additionally, every application that is submitted is safely kept, giving officers access to a searchable and filterable history that they can use for audits or future planning. This module increases transparency and speeds up processing by eliminating the paperwork, preventing delays, and maintaining digital records of all applications and actions taken. Additionally, it guarantees that permit workflows are responsible, visible, and follow council regulations, which boosts overall governance and public trust.

3. Online Payments & Finance Management

Automating financial transactions between the Urban Council and its citizens depends on the Online Payments and Finance Management module. Through a web based interface connected to a trustworthy third party payment gateway, this module enables locals to safely pay for a variety of services, such as taxes, fines, business permits, and building rentals. **GovPay [1]** is the newly introduced payment interface to do payments to government institutes. Even people with no technical expertise can easily complete payments with minimal assistance due to the interface's simplicity and ease of operation. The system automatically creates a digital receipt for users' records after a payment is submitted, and they can download or receive via email.

A real-time dashboard that tracks all incoming payments by category is available to finance officers and system administrators on the administrative side. This makes it possible to track income quickly and easily and aids in early identification of any mistakes. This module's ability to generate financial reports with visuals like pie charts that break down revenue sources by category, including taxes, license fees, and fines, is one of its key features. These reports give the finance department clear documentation for audits and aid in budget planning. The system minimizes manual handling, cuts down on delays, and lowers the possibility of fraud or mistakes by digitizing the entire payment process. Additionally, it ensures that financial operations are traceable, effective, and accessible around-the-clock, enhancing the council's revenue management system and making.

4. HR & Payroll Management

Through the digitization and automation of employee-related processes, the HR and Payroll Management module is intended to support the Horana Urban Council's internal administrative needs. The system allows HR personnel to create and maintain thorough employee profiles that include each worker's department, job title, pay scale, and work history. For a large workforce, this centralized database makes it simple to access and update staff records. The system also manages leave, allowing HR managers to keep tabs on trends over time, approve or deny leave requests, and track leave balances. This module's automated salary calculation is one of its most potent features. The system takes into consideration a number of factors, including initial salary, allowed leave, overtime, bonuses, and deductions.

A possibility of mistakes being made is significantly reduced since these computations take place towards consequently at the end of each pay process. Transparency and self-service options are enhanced by allowing employees to safely log in and view their pay history, leave balances, and salary slips. Employees no longer have to search for usual data in individual, which reduces the workload for the HR department. Reports on payroll are generated as well by the system and can be used for government audits or

internal analysis. In addition to reducing both money and time, this module guarantees compliance, accuracy, and satisfaction for council employees and HR staff by optimizing payroll and HR procedures.

5. Inventory Management

The Inventory Management module is essential for keeping track of the Urban Council's physical assets and supplies, such as office materials, cleaning equipment, maintenance tools, and other public-use items. This module allows authorized inventory officers to add new items to the system, update quantities, assign items to departments, and record usage history. Each item entry includes key details such as name, category, stock level, purchase date, and current status. The system continuously monitors stock levels and provides automatic alerts when certain items fall below the predefined threshold, allowing timely reordering and preventing stockouts. This proactive management ensures that all departments have access to the resources they need to function efficiently. Additionally, the system supports the generation of audit-ready inventory reports, which include summaries of stock movements, item usage by department, and purchasing history. This information is useful not only for routine operations but also for financial planning, budgeting, and regulatory compliance. Furthermore, inventory data can be integrated with the finance module to reflect expenses and with HR to track issued items to staff. The system also supports role-based access control, ensuring that only authorized users can make updates or access sensitive information. Overall, the Inventory Management module brings structure, visibility, and accountability to the council's asset management process, reducing the risk of misuse or loss while improving overall operational readiness.

6. User Management

The User Management module is the backbone of the **Horana Urban Council Smart Management System**, ensuring that all system interactions are secure, organized, and

role-specific. This module allows administrators to create, update, and manage user accounts for different roles, including citizens, officers, HR staff, inventory officers, finance officers, and system admins. Each role is assigned specific permissions based on their responsibilities, ensuring that users can only access the modules and data relevant to their duties. For example, citizens can submit complaints, apply for permits, and make payments, but they cannot view internal HR or inventory data. Officers can review and act on citizen requests, while admins have full access to manage all modules and oversee system activity. The system also includes secure login functionality with email-based authentication to verify user identity, reducing the risk of unauthorized access. Passwords are encrypted, and login sessions are monitored for security purposes. Additionally, automated email notifications are integrated into this module to inform users of account creations, password changes, or important system updates. The User Management module also supports account deactivation for inactive users and provides an audit log to track user activities within the system. This log helps maintain accountability and transparency in all operations. By implementing structured role-based access control, the module ensures data protection, minimizes errors, and improves overall operational efficiency. It forms the foundation for secure, controlled, and organized system usage across all departments and user groups in the Horana Urban Council.

Non-Functional Requirements

❖ Usability

- The system will be designed with a clean and intuitive user interface to ensure a smooth user experience for both technical and non-technical users.
- Forms and dashboards will be simple, with clear instructions and minimal steps required to complete any action.
- Citizens, officers, and employees will be able to use the system with minimal training.

❖ Availability

- The platform will be available **24/7**, allowing users to access services at any time, from any location, using internet-enabled devices.
- Cloud-based deployment will help maintain high uptime and ensure continuous availability.

❖ Performance

- The system responds quickly to user interactions and processes tasks with minimal delay. Fast response times
- contribute to a positive user experience, allowing users to navigate through the vendor management portal
- smoothly and complete their tasks without unnecessary delays

❖ Security

- The system should enforce robust security measures, ensuring that only authorized personnel have access to specific functionalities and that data is transmitted securely.
- Protect user data and sensitive information from unauthorized access or attack.
- Safe storage of payment and personal information.
- Multi-factor authentication limits administrator access.

❖ Maintainability

- The system will be developed using clean, modular code with proper documentation.
- Git and GitHub will be used for version control, enabling smooth collaboration, rollback, and tracking of changes.

- Future updates, bug fixes, or new feature implementations can be easily deployed without disrupting existing services.

Technical Requirements

❖ Hardware Requirements

The system must be deployed on hardware that meets minimum specifications to ensure smooth performance, especially when handling multiple simultaneous users and processing large volumes of data.

❖ Frontend

- React.js - for building fast, dynamic, and user-friendly interfaces.

❖ Backend

- Node.js with Express.js - for handling server-side logic and REST API development.

❖ APIs and Integrations

- RESTful APIs – for seamless communication between frontend and backend.
- Third-party Payment Gateway – for secure and real-time transaction processing.
- Chatbot Integration – for guiding users or answering FAQs automatically.

❖ Database

The system will use **MongoDB** as the database management system. This database is chosen for its scalability and flexibility, which is ideal for storing different data types,

1. Citizen complaints and service requests
2. Permit and license applications
3. Payment and financial records
4. Employee profiles and payroll information
5. Inventory data

MongoDB ensures fast read/write operations, proper indexing, and minimizes the risk of data loss through replication and backup options.

❖ User Interface (UI)

The system will feature a user-friendly, responsive web interface designed using React.js. The UI will be, Easy to navigate for both citizens and staff with minimal technical knowledge. Designed with clean layouts, dropdowns, filters, notifications, and clearly labeled buttons for minimal user experience. Each user role will see a customized dashboard relied on their permissions. Developed in three languages to easier to citizen.

❖ Notifications

Email Notifications: The system will include email-based notifications to alert users of key actions such as account registration, application status updates, successful payments, and password reset requests. This helps enhance user awareness and detect any suspicious activity.

❖ Hosting & Deployment

Cloud-hosted using platforms like Render, Vencel, or AWS to ensure uptime, scalability, and easy deployment.

❖ Version Control

Managed using Git and GitHub for team collaboration and version tracking.

Literature Review

To get better understand how our proposed system can improve local government services, we looked at how other councils in Sri Lanka are currently using technology. One of the most examples is the **Colombo Municipal Council [2]**, which provides basic online services through its website.

when we looked more closely, we noticed that the system is mainly focused on online payments. There are no features for real time services, submitting complaints online, applying for permits or licenses, managing staff records, or tracking inventory. Most of those tasks still happen through manual, paper-based processes. Also, the website itself has a very simple interface, and it may not be easy for everyone to use especially those who are less familiar with online systems.

We also considered the example of the **Gampaha Municipal Council [4]**, which recently introduced a more advanced payment system. Citizens there can now pay for services like cremation bookings or hall rentals using a digital payment gateway. While this is definitely a step forward, it's still limited to payments only. Like Colombo, Gampaha's system doesn't yet support a full digital workflow for other council services. That's where our system stands out.

The **Horana Urban Council Smart Management System** will go far beyond just handling payments. It will be designed to fully digitalize the council's operations from complaint handling and permit processing to payroll, leave tracking, and inventory management. It also includes secure login, user role management, real time notifications, and a clean interface that citizens and staff easier to use. Finally, our system will be built to fill the gaps left by current solutions, helping the council deliver faster, more transparent, and more efficient services to the community.

Methodology

To successfully design and implement the Horana Urban Council Smart Management System, we followed a structured development methodology that ensured efficiency, collaboration, and flexibility throughout the project. Our approach includes five key phases: Requirements Engineering, System Design, Development, Testing, and Deployment all aligned under the Agile development methodology.

Agile Methodology

Agile Methodology is a way to manage projects by breaking them into smaller parts. It focuses on working together and making constant improvements. Teams plan, work on the project, and then review how things are going in a repeating cycle.[5]

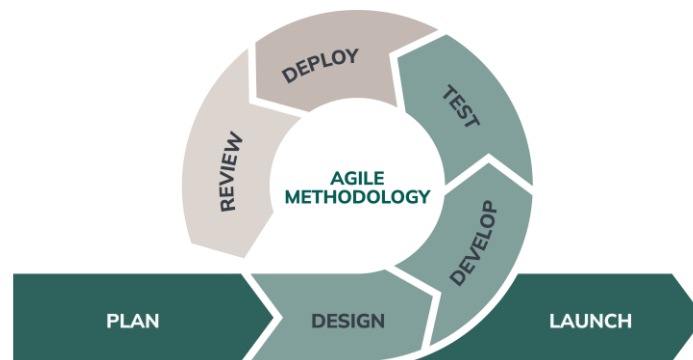


Figure 1

We Selected the Agile methodology as it lets us to build the system in manageable sprints, respond to changing requirements, and get continuous feedback from users. Agile fits our project well since we are developing multiple modules, like Citizen Requests, Payments, and HR, that can be built and tested separately in sprints. Each sprint focused to deliver a working part of the system. This was followed by internal reviews, bug fixes, and improvements. We held stand up meetings to monitor progress and reassign tasks as needed.

Requirement Gathering

Requirement Gathering In this stage, our project team met with key stakeholders at the Horana Urban Council, including officers, admin staff, HR officers, and a few citizens, to understand the challenges they face with the current manual system.

Activities,

- Identifying the pain points in current complaint handling, payroll, licensing, and inventory processes.
- Documenting detailed functional requirements (ability to submit complaints online, auto calculate salaries)
- Defining project scope, main modules, and goals. Outlining timelines, available resources, and initial budget estimations.
- Creating a project plan with task allocation and sprint scheduling.

Design

In this phase, we planned how the system would work both technically and visually.

- Designing the overall system architecture using UML diagrams (Use Case, Activity, and Class diagrams).
- Building wireframes and mock user interfaces using Figma to visualize key pages.
- Finalizing how modules interact (how payments are linked to license approval or inventory updates).

Development Tools and Technologies





Structuring and managing data in MongoDB for entities like complaints, users, payments, and licenses. MongoDB simplifies data management with its intuitive **document model**, mapping unique objects to distinct documents.[6]



Express.js. - Express is a minimal and flexible



Designing responsive and interactive UIs using React.js. React will be used to create a responsive and interactive front end for our web application [7]



Writing backend APIs using Node.js - Node JS is a single-threaded, open-source, cross-platform runtime environment for building fast and scalable server-side and networking applications [8]

- Implementing login authentication and role-based access control.

Apart from MERN Stack we use the following tools and technologies in our project.



Visual Studio Code is a streamlined code editor with support for development operations like debugging, task running, and version control



GitHub is a code hosting platform for version control and collaboration. It lets us to work together on projects from anywhere.

Communication Tool

- Microsoft Teams: For conducting daily scrum meetings Sprint planning, Microsoft Teams will be used and it ensures effective communication among members.
- WhatsApp: A WhatsApp channel with all members has been created to stay frequently connected to the development of the web application.

Testing

This phase involves several types of testing:

- Integration Testing: Ensuring that different components work together.
- System Testing: Testing the entire system as a whole.
- User Acceptance Testing: Confirming that the software meets user requirements.
- Performance Testing: Assessing the system's speed, scalability, and stability.

Integration method

The Horana Urban Council Smart Management System uses RESTful APIs to ensure efficient communication between system modules and components. By using RESTful APIs, the system maintains modular design, scalability, and security, supporting a responsive and efficient web application.

Examples of API usage within the system include:

- Transferring citizen complaint data to relevant officer dashboards.
- Linking permit/license applications with the payment module for real-time status updates.
- Fetching employee leave and overtime records for accurate payroll processing.
- Syncing inventory changes with the finance module for reporting.

Additionally, APIs are used for:

- Sending automated email notifications to users.
- Managing role-based access control to ensure secure access.
- Enabling future integrations such as payment gateways and multi-language support (e.g., Sinhala, Tamil).

Project Plan

The project will use an Agile Scrum approach where the work is organized into several 2-week sprints. This method allows the team to deliver the system in smaller, functional parts instead of waiting until the end of development for a complete release. Each sprint will focus on specific deliverables, making sure that progress is clear and measurable. This approach supports flexibility; the team can change the system's features based on ongoing feedback from stakeholders, which improves both quality and relevance to real-world needs.

High-priority modules like Citizen Service Requests, Permit and License Management, and Online Payments will be developed in the first sprints. This ensures that the Urban Council can start benefiting from key functionalities quickly, even before the complete system is finished. Weekly stand-up meetings, sprint reviews, and backlog grooming sessions will keep the team on track. Continuous integration and deployment practices will help maintain quality throughout development.

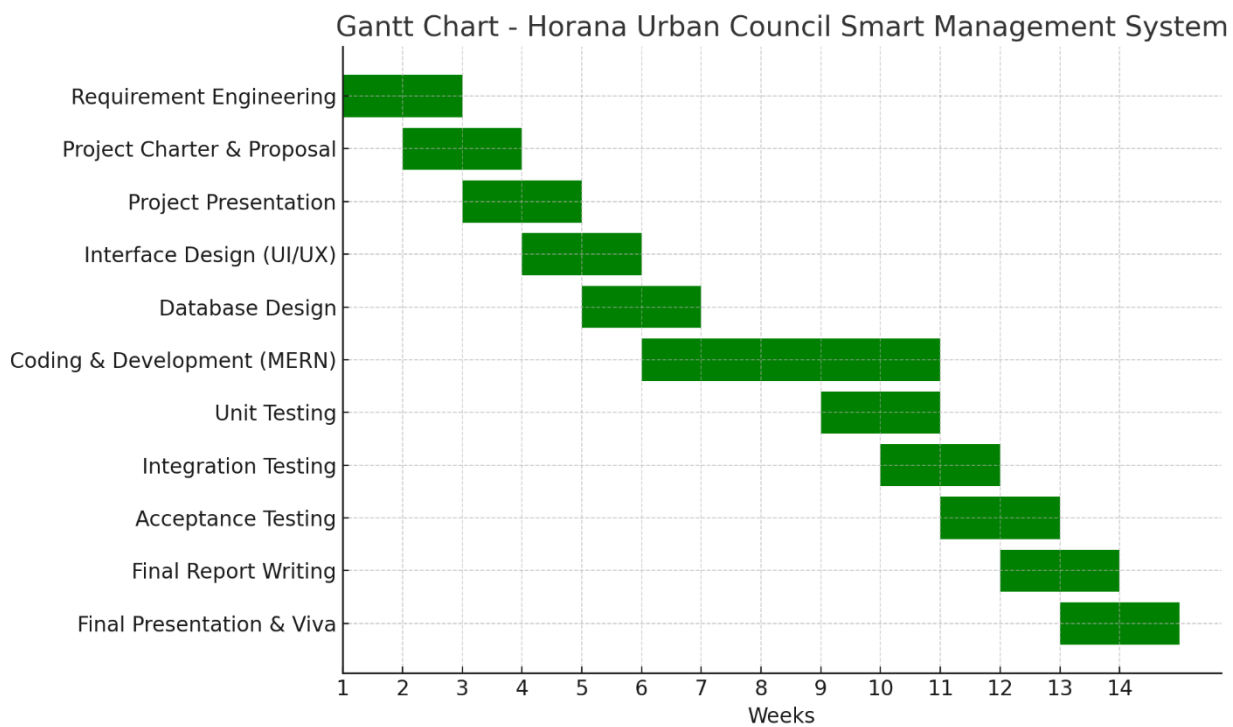


Figure 2

Work Breakdown Structure

Task Name	Description
Requirement Gathering	Gather detailed requirements from Horana Urban Council staff, including Citizen Service Officers, HR staff, Finance officers, and Inventory managers. Use surveys and interviews to document service processes and user needs.
Project Proposal	Prepare initial proposal, define scope, objectives, and methodology.
Project Presentation	Present project plan to supervisor/client for feedback.
Interface Design (UI/UX)	Implement online application forms, approval workflows, and email notifications
Database Design	Create MongoDB schema for all entities (Users, Requests, Payments, Inventory).
Coding & Development (MERN)	Develop backend APIs (Node.js, Express.js) and frontend components (React.js). Integrate payment gateway and role-based authentication.
Unit Testing	Test individual modules for correctness (Citizen Requests, Payments, etc.).
Integration Testing	Get feedback from actual Horana Urban Council staff and make adjustments.
Acceptance Testing	API connections between modules (e.g., payments, email notifications)
Final Report Writing	Compile final documentation, user manuals, and technical guides.

Final Presentation & Viva	Present completed system to supervisor, examiners, and stakeholders.
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Table 1

Evaluation Method

1. Functionality Testing (CRUD Operations)

This testing ensures all basic functions of the system work correctly. For each module:

- **Citizen Service Requests:** We will check that citizens can log complaints, officers can update the status of these complaints accurately, and administrators can identify and delete any duplicate entries to keep data accurate.
- **License & Permit System:** Testing will confirm that citizens can submit applications, officers can approve or reject them with relevant comments, and citizens receive timely notifications, including email alerts to keep them informed about the application status.
- **Finance Module:** We will test the system to ensure online payments are processed correctly, users can download receipts, financial reports are generated accurately, and admins have access to these reports. Users will also receive email confirmations upon successful payment transactions.
- **Salary & HR:** Testing will verify the accurate management of salary data, correct tracking of employee leaves balances, and proper payroll calculations that consider overtime and leave deductions.
- **Inventory Management:** We will check that staff can add new inventory items, update stock levels as needed, generate reports reflecting current stock, and that alerts are triggered when stock levels fall below set thresholds.

2. User Testing by Actual Staff

Involving real users like citizens, officers, HR, and finance staff helps us evaluate the system's usability and ease of use. This approach allows us to identify any challenges or issues they face in their daily tasks, letting us gather useful feedback for improving the system design and workflows to ensure it is user-friendly and effective.

3. Admin Verification of Reports

Administrative users will review all generated reports financial, HR, and inventory to ensure that:

- Data presented is accurate and matches underlying records.
- Reports are complete and clearly formatted for easy understanding.
- Features like filtering data by criteria and exporting reports to other formats (like PDF or Excel) work correctly without errors.

4. Performance Benchmarks

Performance testing measures how quickly and reliably the system responds to user actions. Key tasks, such as complaint submissions, license approvals, payment processing, salary updates, and inventory queries will be timed to ensure they meet acceptable standards. We will test the system under both normal and peak loads to confirm it remains stable and responsive even when many users are active at the same time.

5. Security Audits (Login, Role Access)

Security audits will check how strong the system's login and authentication mechanisms are to prevent unauthorized access. We will verify role-based access controls to ensure users can only access features and data suitable for their roles, which helps prevent privilege escalation. Additionally, we will test the system for common vulnerabilities, such as unauthorized data access and injection attacks, to protect sensitive information and maintain overall system security.

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Appendix

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